## ☺ Ground Segment

- Plan end-of-life disposal — ACT-14
- Monitor and command transport — ACT-13

CE-14 TT&C uplink

CE-15 TT&C downlink

## Cislunar Cargo Tug

### Electrical power — LC-2 PWR

- Generate electrical power — SF-03.5
- Store and distribute electrical power — SF-03.6
- Discharge and isolate batteries — LF-06

*100 V bus · 29.1 kW EOL · 8.8 kWh*

CE-05 Power commands

### Command and data handling — LC-5 CDH

- Perform disposal manoeuvre — SF-07.1
- Validate and route commands — LF-02
- Manage modes and failures — SF-03.9
- Collect and format telemetry — LF-03

CE-08 Received messages

CE-09 Telemetry frames

### Communications — LC-6 COMMS

- Receive radio messages — LF-01
- Transmit telemetry — LF-04

CE-16 Proximity link

CE-04 Propulsion commands

CE-06 GNC commands

### Electric propulsion — LC-1 PROP

- Generate thrust — SF-03.1
- Store and feed propellant — SF-03.2
- Vent residual propellant — LF-05
- Provide proximity thrust — LF-09

*5+1 strings · 25 kW · ≥ 3,400 kg Xe*

CE-01 High-power bus

CE-03 Thrust commands

### Guidance, navigation and control — LC-3 GNC

- Determine orbit and attitude — SF-03.3
- Control trajectory and attitude — SF-03.4
- Rendezvous and proximity navigation — SF-03.10

### Thermal control — LC-7 TCS

- Control temperatures and reject heat — LF-07

*3.3 kW*

CE-10 Thermal interface

CE-02 Xenon transfer line

CE-07 Proximity state

### Docking and refuelling interface — LC-4 DRI

- Capture cargo in Earth staging orbit — SF-01
- Receive propellant in orbit — SF-02
- Hand over cargo at lunar destination — SF-04
- Hand over return cargo in Earth orbit — SF-06
- Collect return cargo — SF-05

*1 androgynous port + xenon coupling · 54 matings*

CE-17 Cargo delivery

CE-18 Return cargo

### Structure — LC-8 STR

- Carry loads and launcher interface — LF-08

Logical System

CE-19 Returned cargo

## ☺ Sun

- Provide sunlight — A-20

CE-20 Sunlight

## ☺ Propellant Module

- Supply propellant in orbit — ACT-18

CE-13 Xenon coupling

## ☺ Launch Vehicle

- Launch cargo and propellant — ACT-04

CE-11 Launch adapter

CE-12 Cargo and module stack

## ☺ Lunar Orbit Destination

- Clear approach and docking — ACT-15
- Receive delivered cargo — ACT-16
- Provide return cargo — ACT-17

## ☺ Cargo Customer

- Receive returned cargo — ACT-03

Power · Xenon · Cargo and mechanical · On-board data · Radio link · Heat